

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25_ajust.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-08 09:55:51

Training Start: 2025-05-08 09:48:15

Training End: 2025-05-08 09:53:45

Training Duration: 5.49 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: elu

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.9279

MSE: 6.3900

RMSE: 2.5278

R2 Score: -0.1129

Multi-Step Forecast:

Forecast Steps: 20

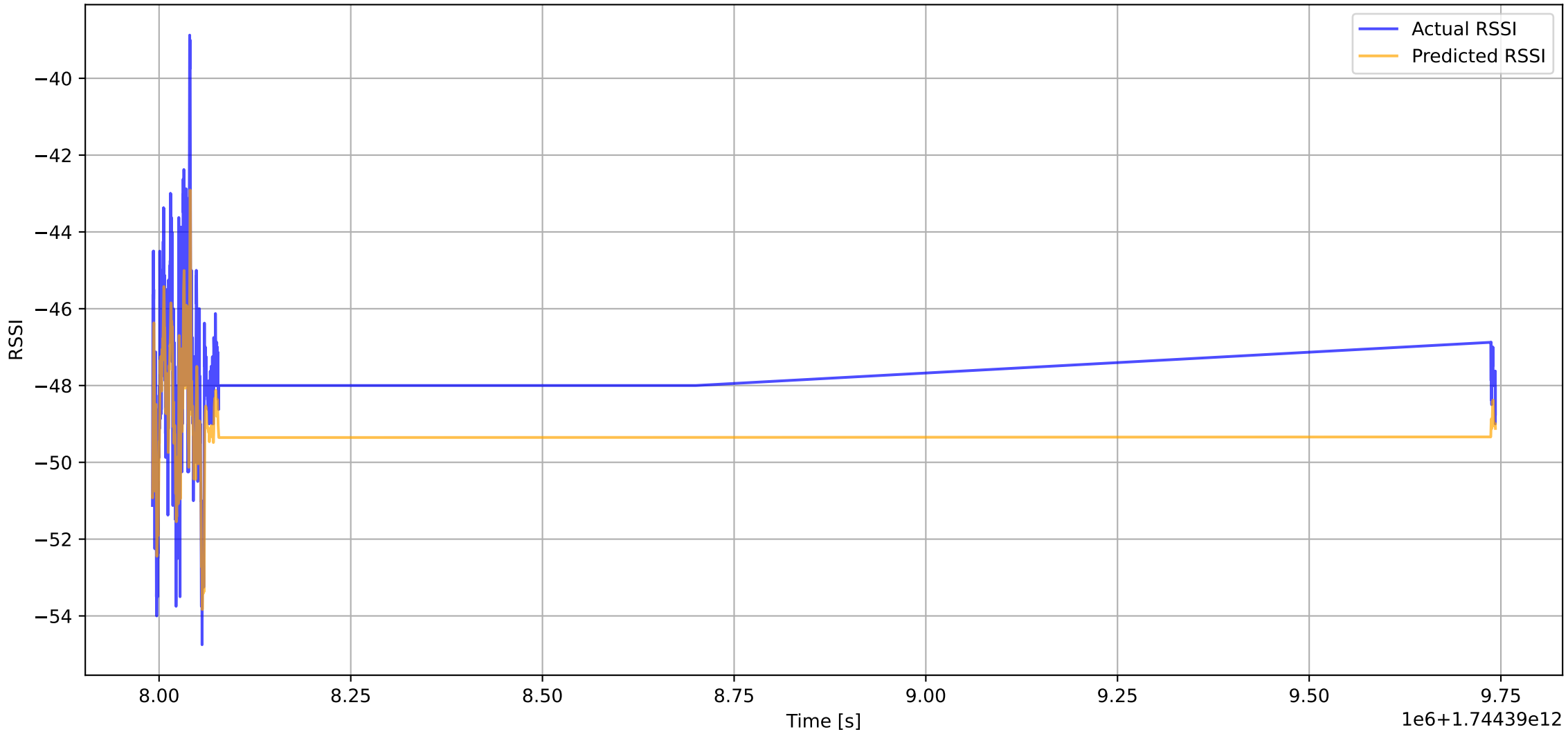
MAE: 0.0520

MSE: 0.0042

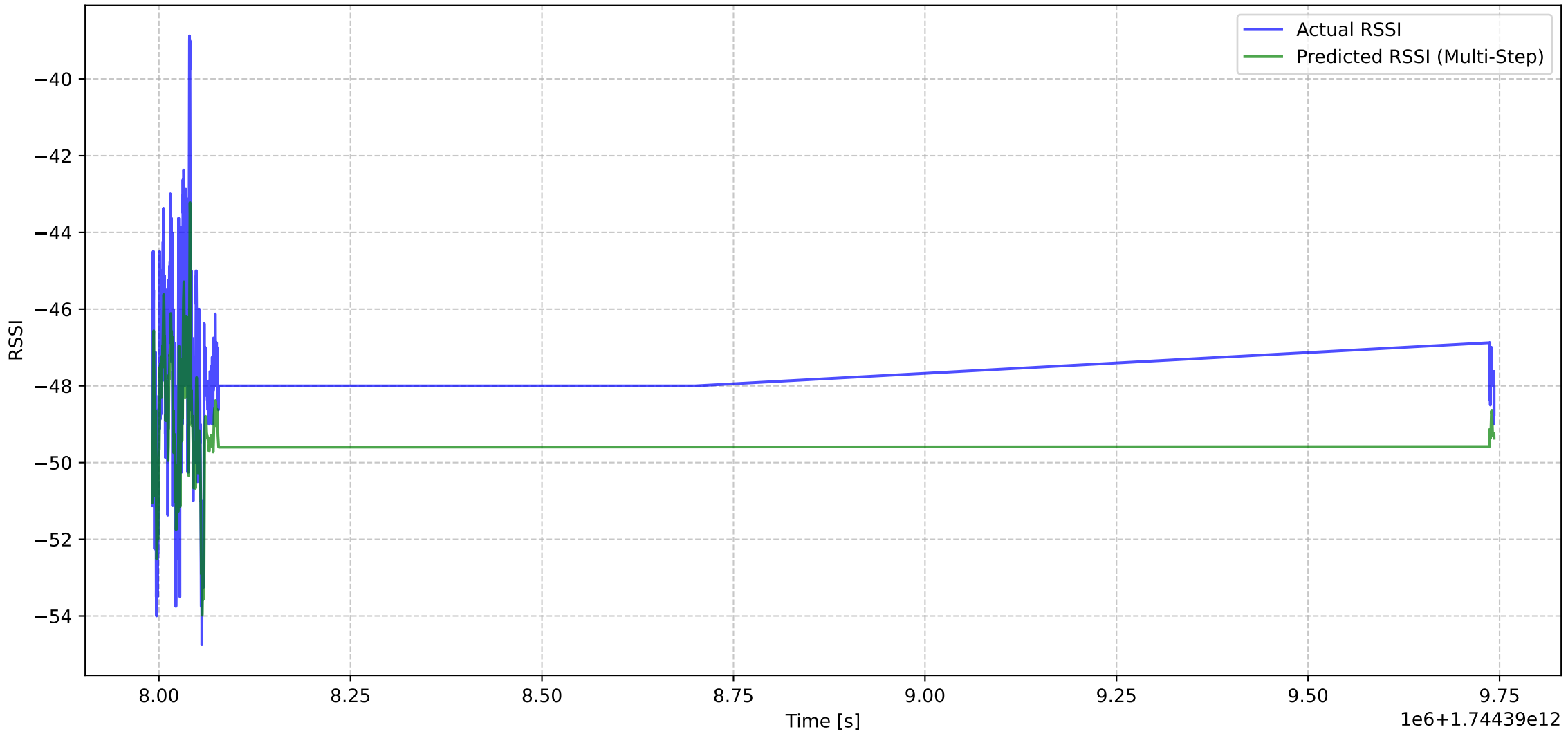
RMSE: 0.0651

R2 Score: 0.4668

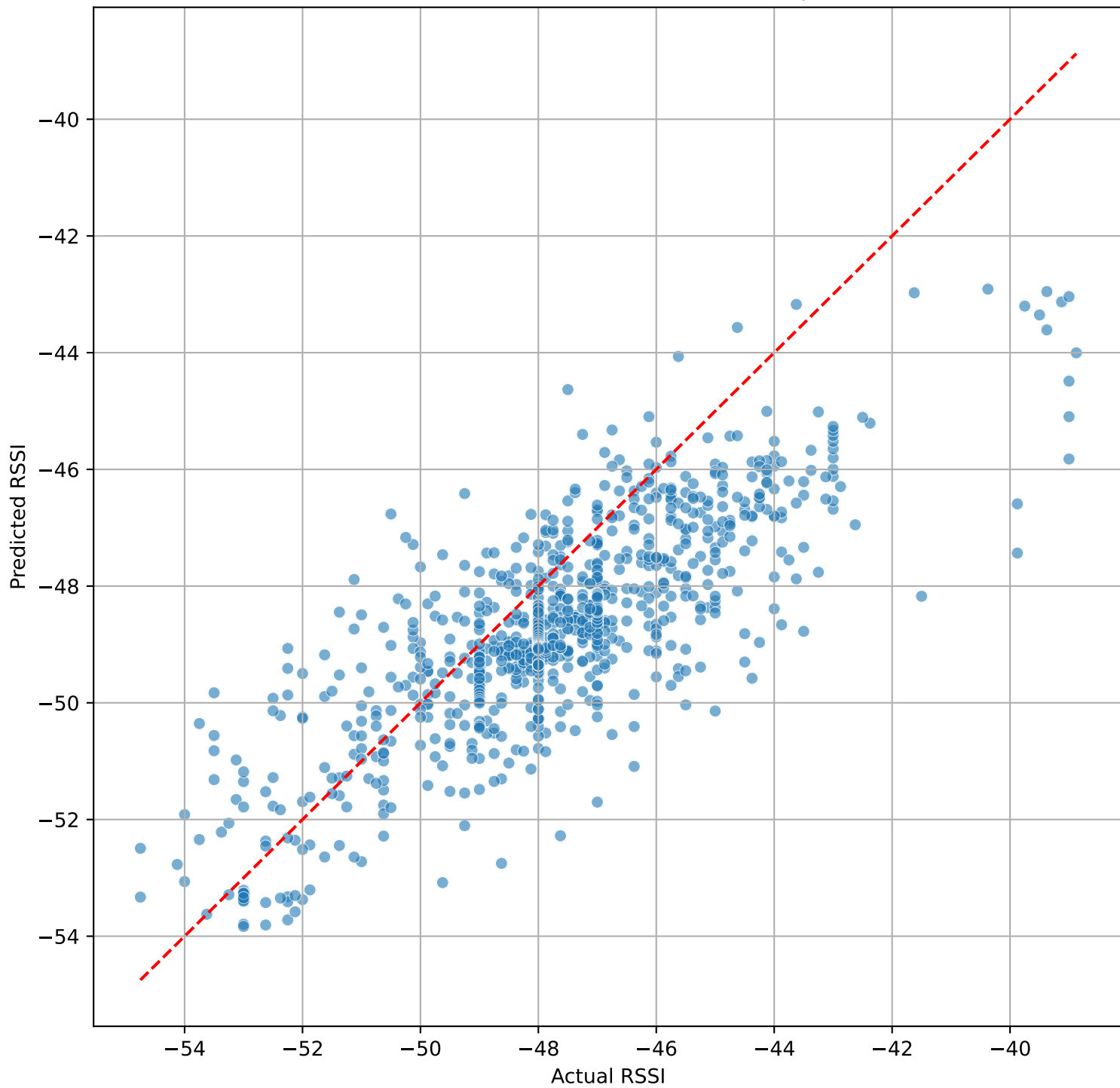
Actual vs Predicted RSSI (First Step)



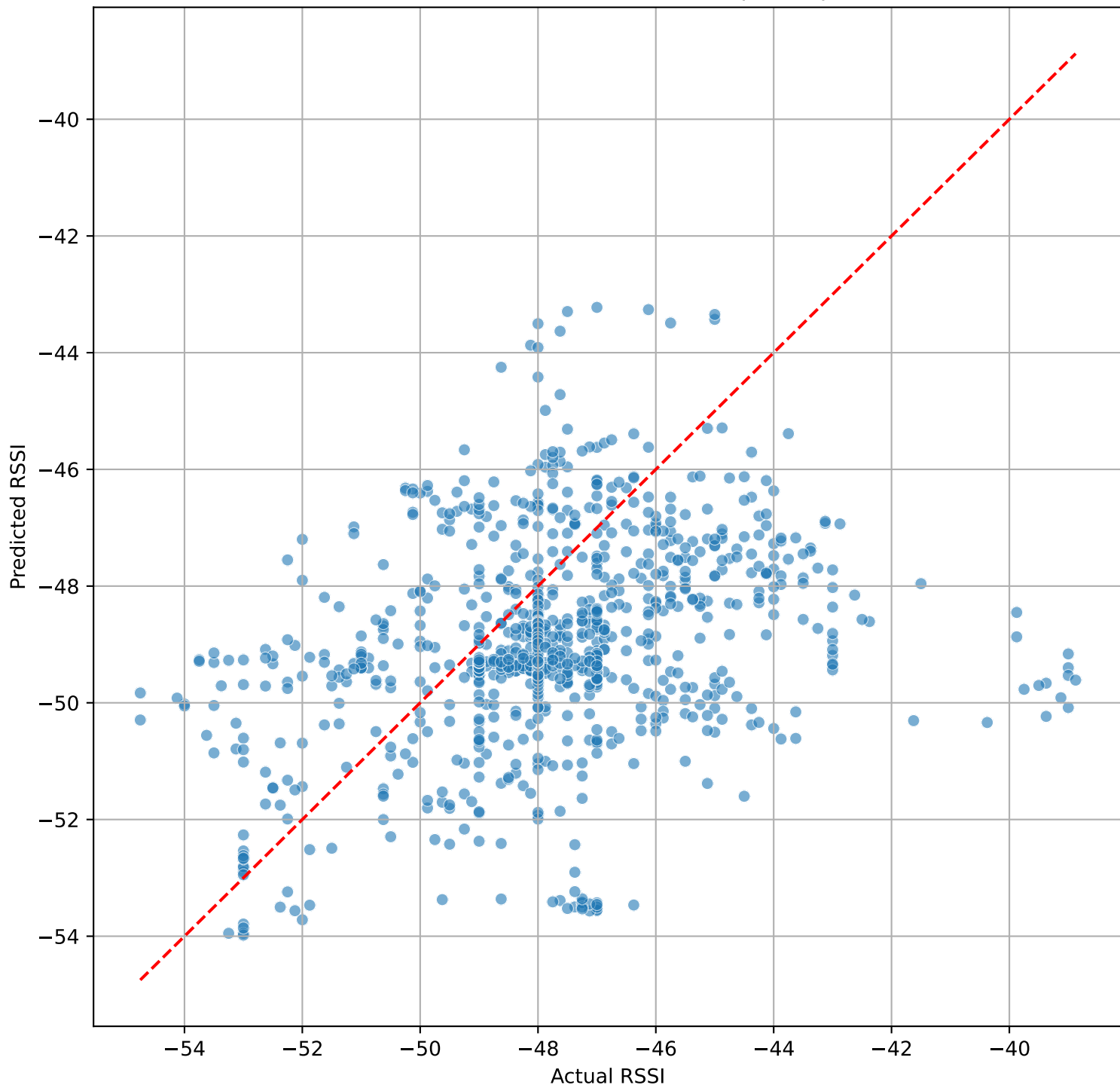
Actual vs Predicted RSSI (Multi-Step (step 20))



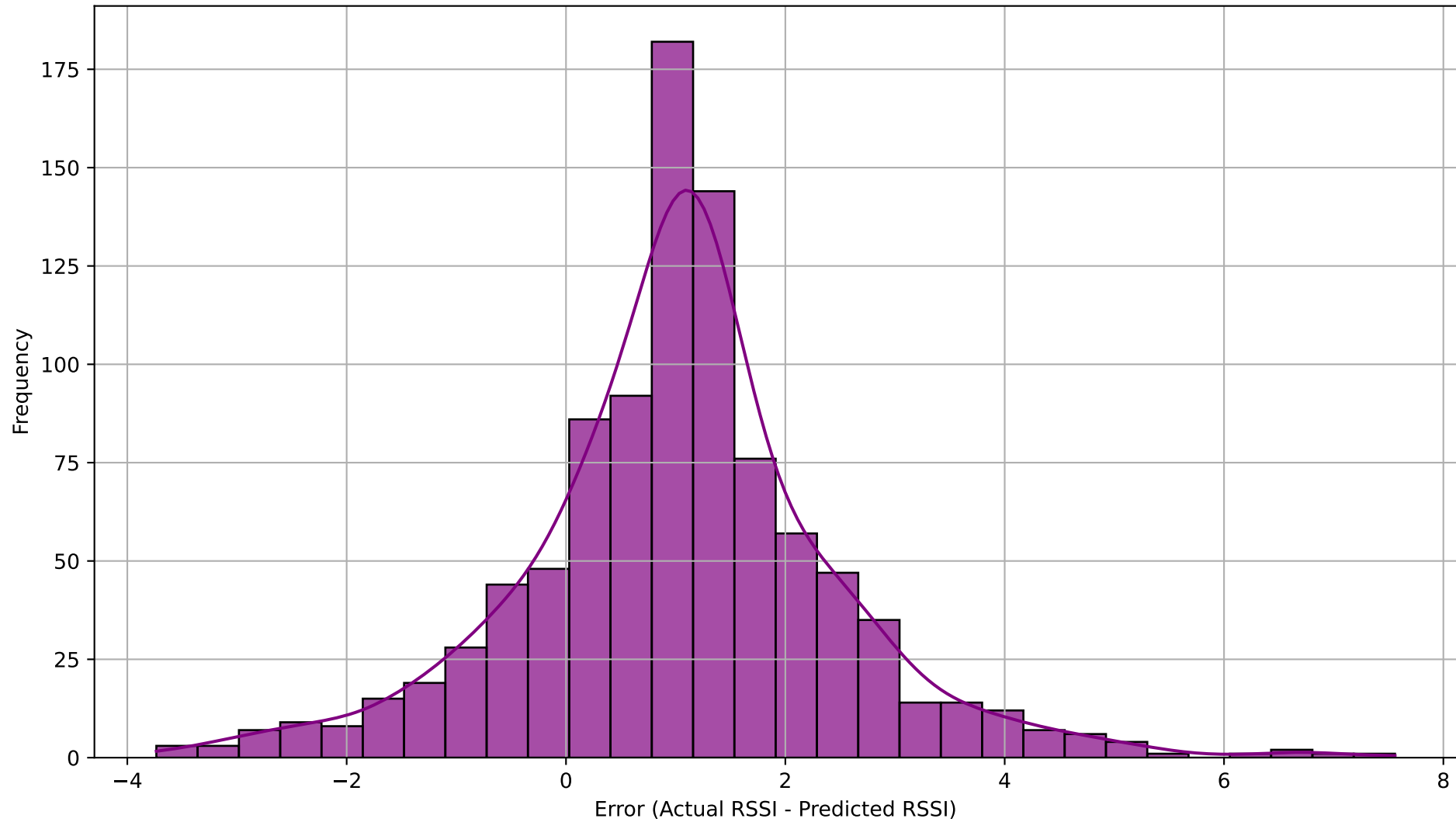
Actual vs Predicted RSSI (First Step)



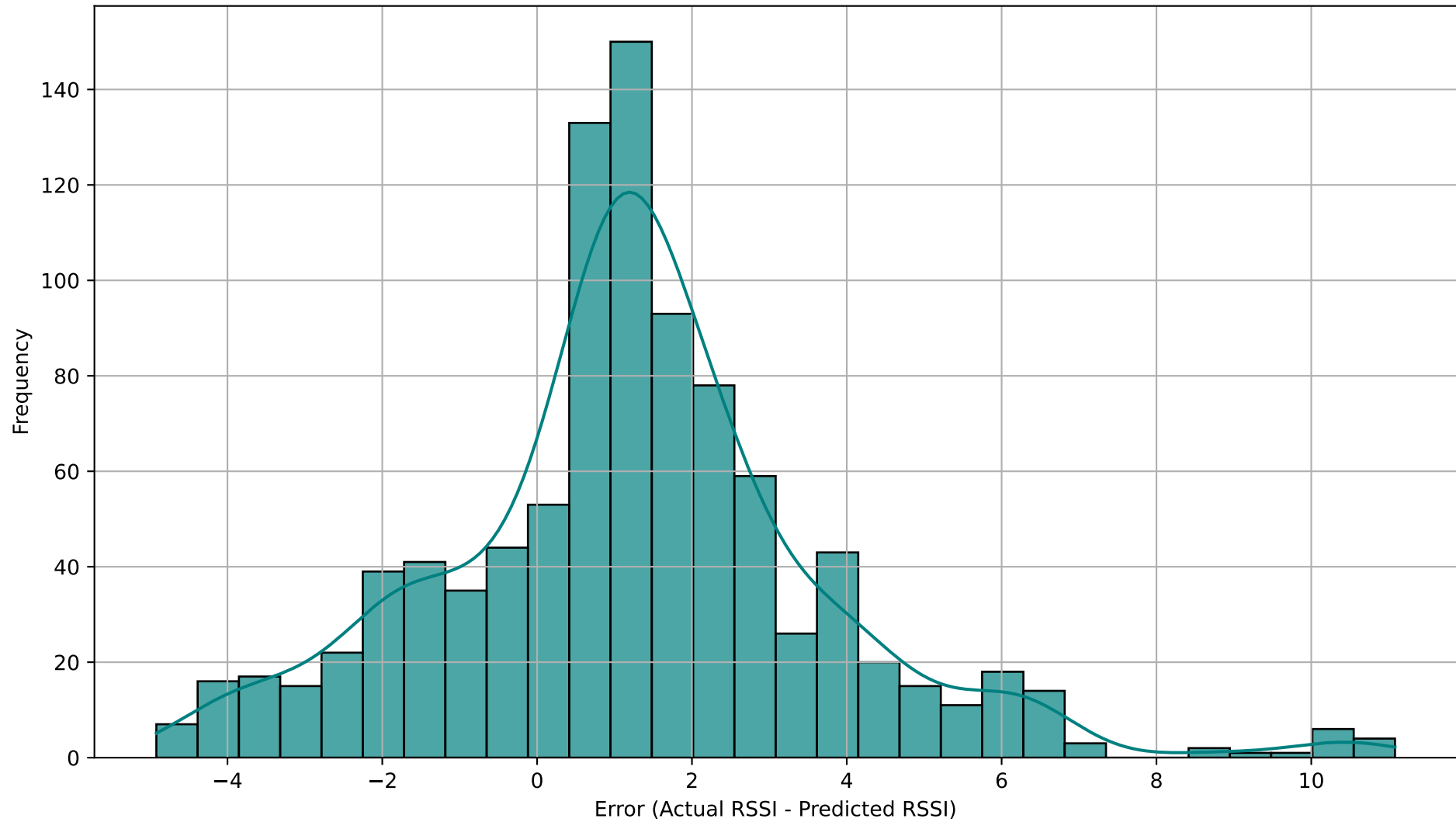
Actual vs Predicted RSSI (Last Step - step 20)



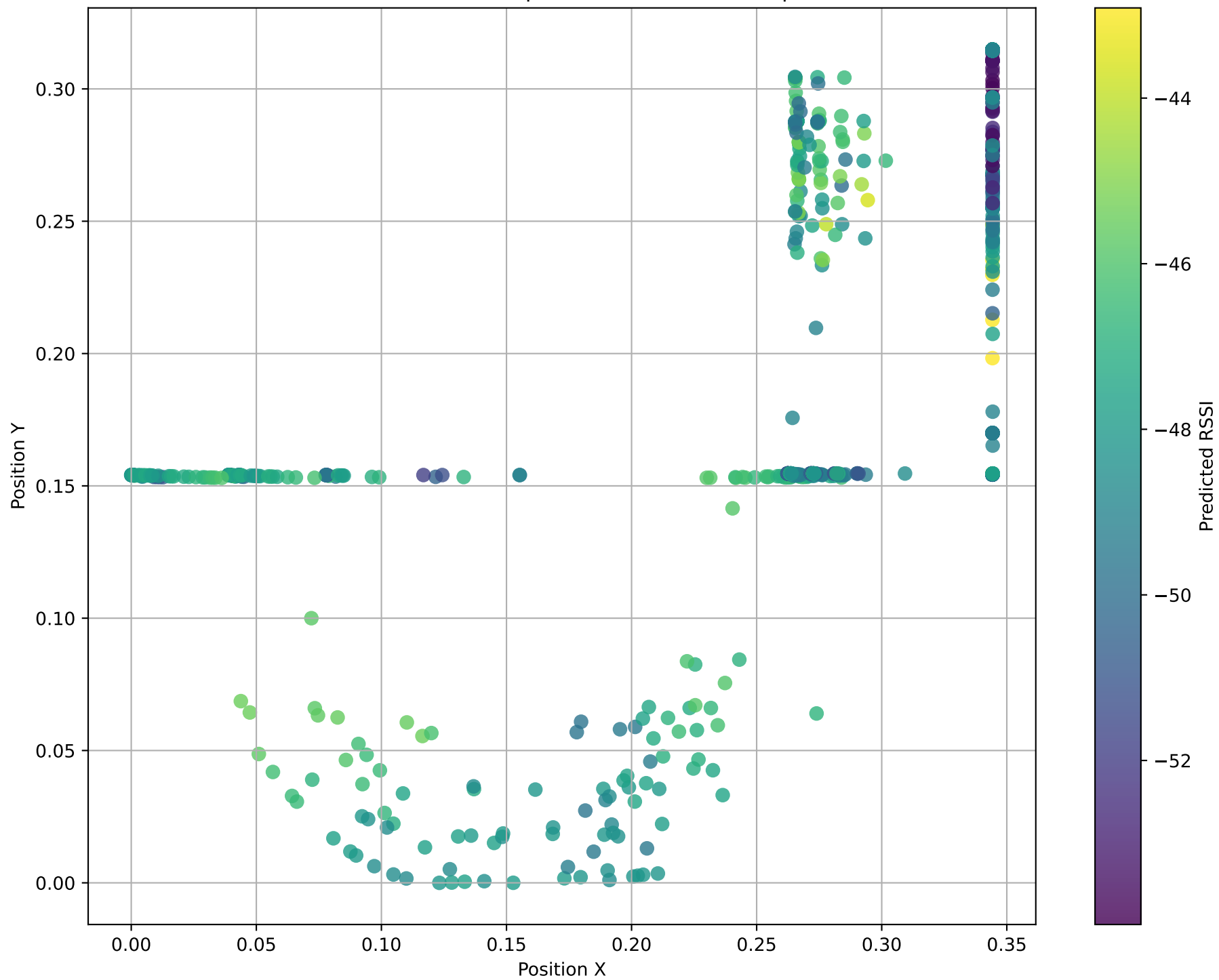
Distribution of Prediction Errors (First Step)



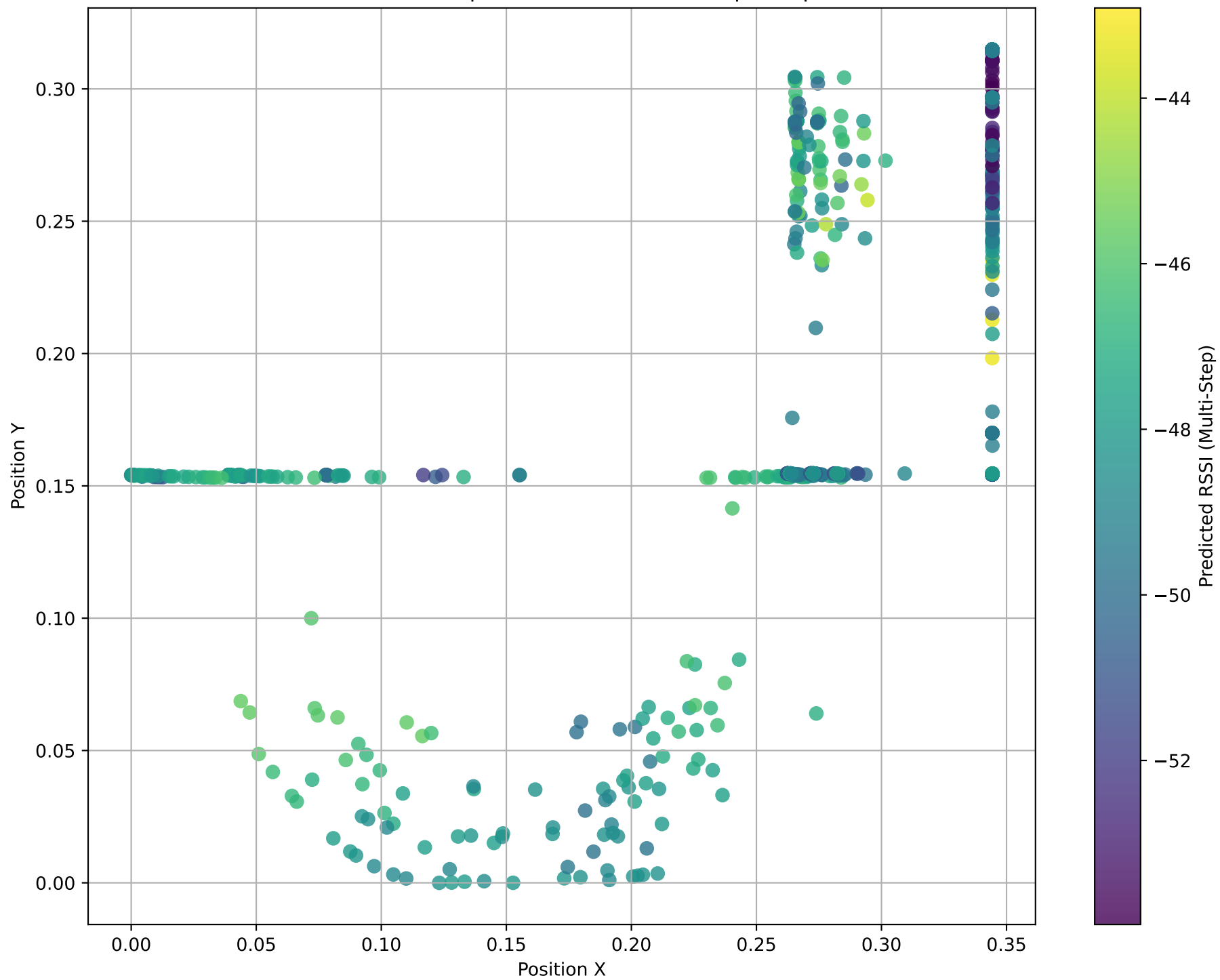
Distribution of Prediction Errors (Last Step - step 20)



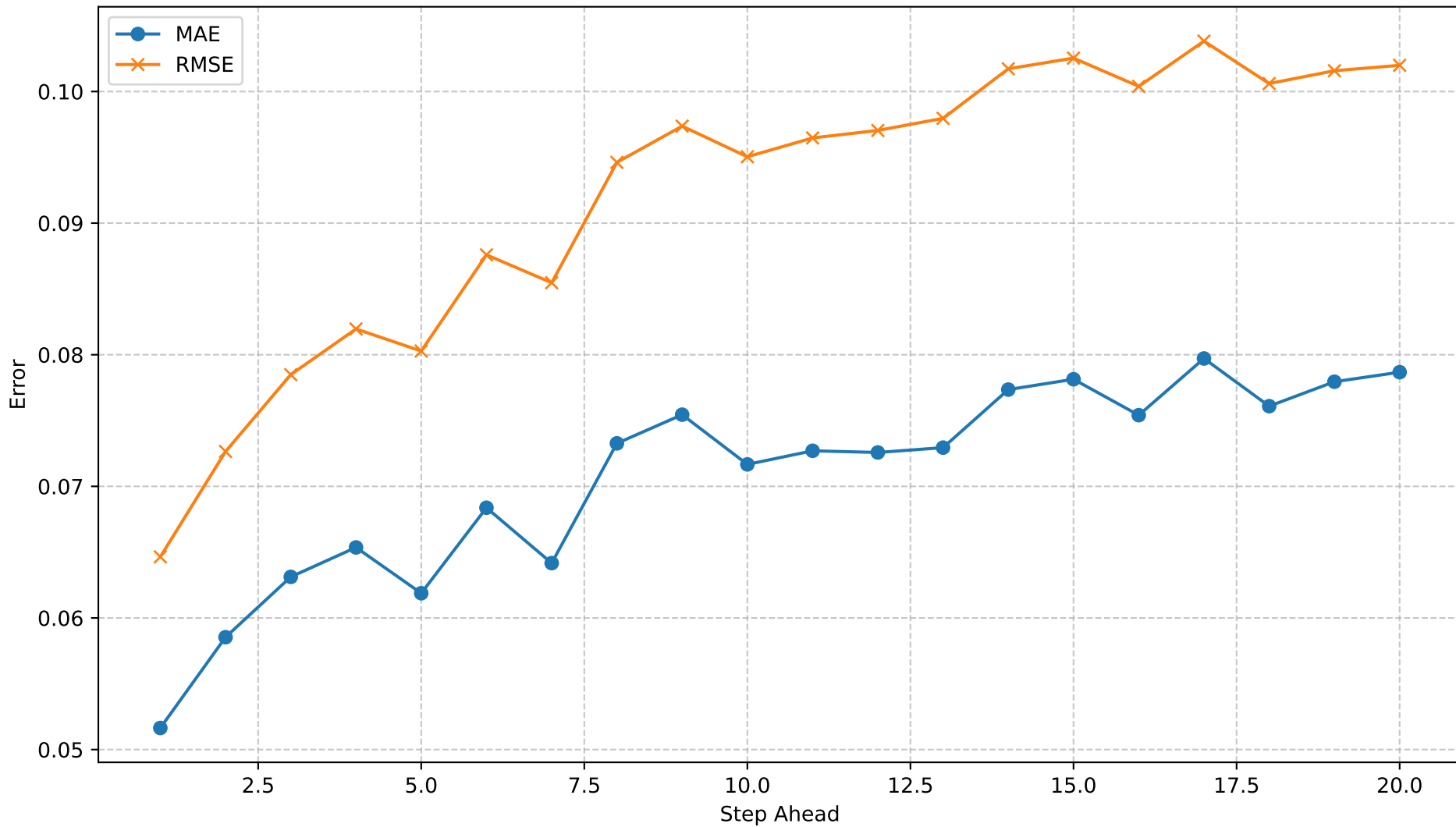
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_11apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.05 / 0.06
- Mediana : 0.07 / 0.10
- 20º passo : 0.08 / 0.10

Crescimento do erro: 52.3% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 17 (0.08), 20 (0.08), 15 (0.08)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.